Cislunar Solar Sailing for DARPA Cattle Run

Simulation study results: transit, launch geometry, cislunar transfers, pole-sitting, 'in-irons' limits, field of view, hybrid agility and the demonstration orbit

Proposal-team briefing | DARPA-SN-27-03 Proposers Day 10 Nov 2026 | 1 November 2026

Bottom line

A hybrid sail flown first at ~1,000 km, then launched toward the lunar plane, meets every Cattle Run objective

1.3 yr
GEO to lunar distance with a Solar Cruiser-class sail (CR-S)

34.5 deg
sunward cone where a sail produces no thrust at all

2.2x
7-day reach of the hybrid vs the sail (from a DRO)

~1,000 km
lowest robust demo orbit (drag < 25 % of sail thrust)

- Baseline a hybrid: Solar Cruiser-class sail + 20 mN two-axis gimballed SEP (CR-H) removes the in-irons region and decouples pointing and power from thrust for 20 kg.
- Demonstrate below GEO at ~1,000 km dawn-dusk SSO (no eclipses, ~20 % faster raising); from a 550 km rideshare, climb with the sail stowed (18 days, 2 kg).
- Launch cislunar elements toward the lunar plane (Cape Canaveral 28.5 deg / 27 deg GTO); DRO, NRHO and L1/L2 halos are weeks apart, L4/L5 two months.
- Do not baseline a lunar pole-sitter: it needs 1.2-1.8 mm/s^2 continuously - an SEP mission. Use paired NRHO spacecraft for polar coverage.

Eight questions, one simulator

Each study question maps to an analysis module, a results file and figures in this deck

- Q1 Transit time from GTO, GEO, TLI, LEO
- Q2 Optimal launch inclination
- Q3 Transfers among DRO, LLO, EML1/2/4/5
- Q4 Lunar pole-sitter requirements
- Q5 'In irons': sail vs hybrid
- Q6 Field of view, fixed vs gimballed bus
- Q7 Hybrid manoeuvrability & acceleration
- Q8 Lowest acceptable demo orbit

Simulator: GPU-batched, validated against flight data

NumPy or PyTorch (CUDA) backend - thousands of trajectories propagated in lock-step

sail optical force model, materials, optimal steering, hybrid allocation

environment Sun/Moon ephemerides, shadows, NRLMSIS drag tables

earth_orbit orbit-averaged spirals (MEE), J2, eclipses, drag, SEP

cr3bp Earth-Moon CR3BP + rotating Sun line, halos, NRHO, DRO

designs mass model calibrated on ACS3 and Solar Cruiser

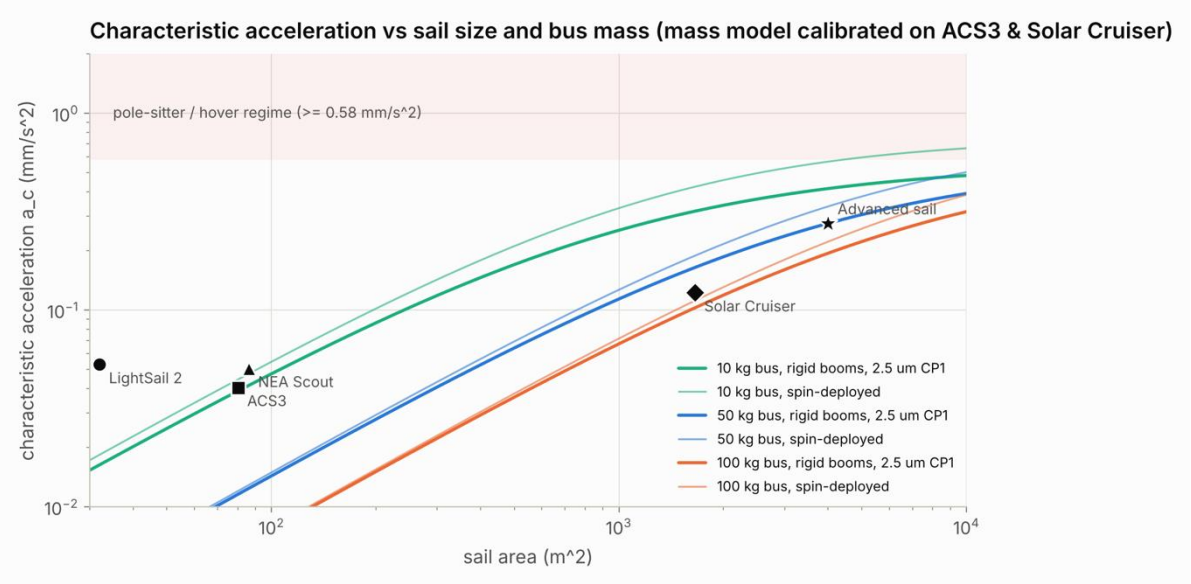
analyses q1-q8 + design trades; run_all.py rebuilds everything

Benchmark	Reference	Simulator
Orbit-averaged thrust fraction	0.4393 (McInnes)	0.4393
ACS3 SMA rise, 900 km, 180 d	+340 km (Dono)	+353 km
Min. ACS3 raising altitude	~691 km (Wilkie)	647 km
Averaged vs Cowell (Sun+Moon), 60 d	-	0.1 % in apogee
9:2 NRHO period / perilune	6.56 d / ~3,250 km	6.56 d / 3,249 km
Lunar pole hover at 60,000 km	Ozimek contours	1.77 mm/s^2

Design points

Heritage sails anchor the range; the study uses a Solar Cruiser-derived sail (CR-S), its hybrid (CR-H) and a far-term sail (CR-A)

Design	Mass (kg)	Area (m^2)	a_c (mm/s^2)	SEP
LightSail 2 (2019)	5	32	0.053	-
ACS3 (2024)	16	80	0.041	-
NEA Scout (2022)	14	86	0.051	-
Solar Cruiser (TRL-6 sail)	111	1,653	0.123	-
CR conventional sail	111	1,653	0.123	-
CR hybrid	131	1,653	0.104	20 mN, 1500 s, +/- 30 deg
Advanced sail (DCB-II, 1 um CP1)	120	4,000	0.275	-

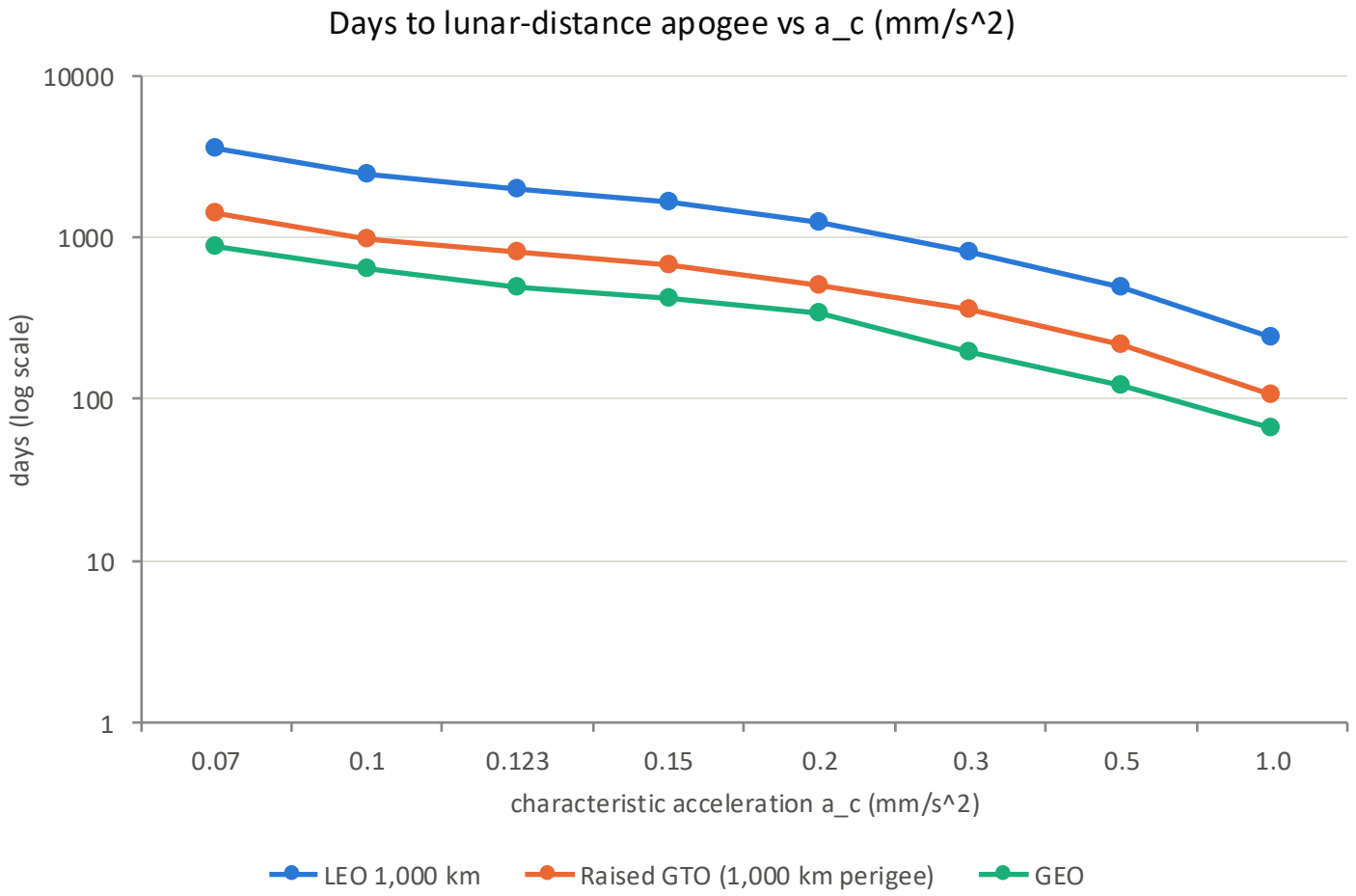


Getting there

Q1 transit time | Q2 launch inclination | Q3 transfers between cislunar orbits

Q1 Transit time scales as 1 / a_c

Near-term sails need 1.3-5.5 years to climb from Earth orbit to lunar distance - start high



490 d

GEO to lunar distance (CR-S)

819 d

raised GTO (1,000 x 35,786 km)

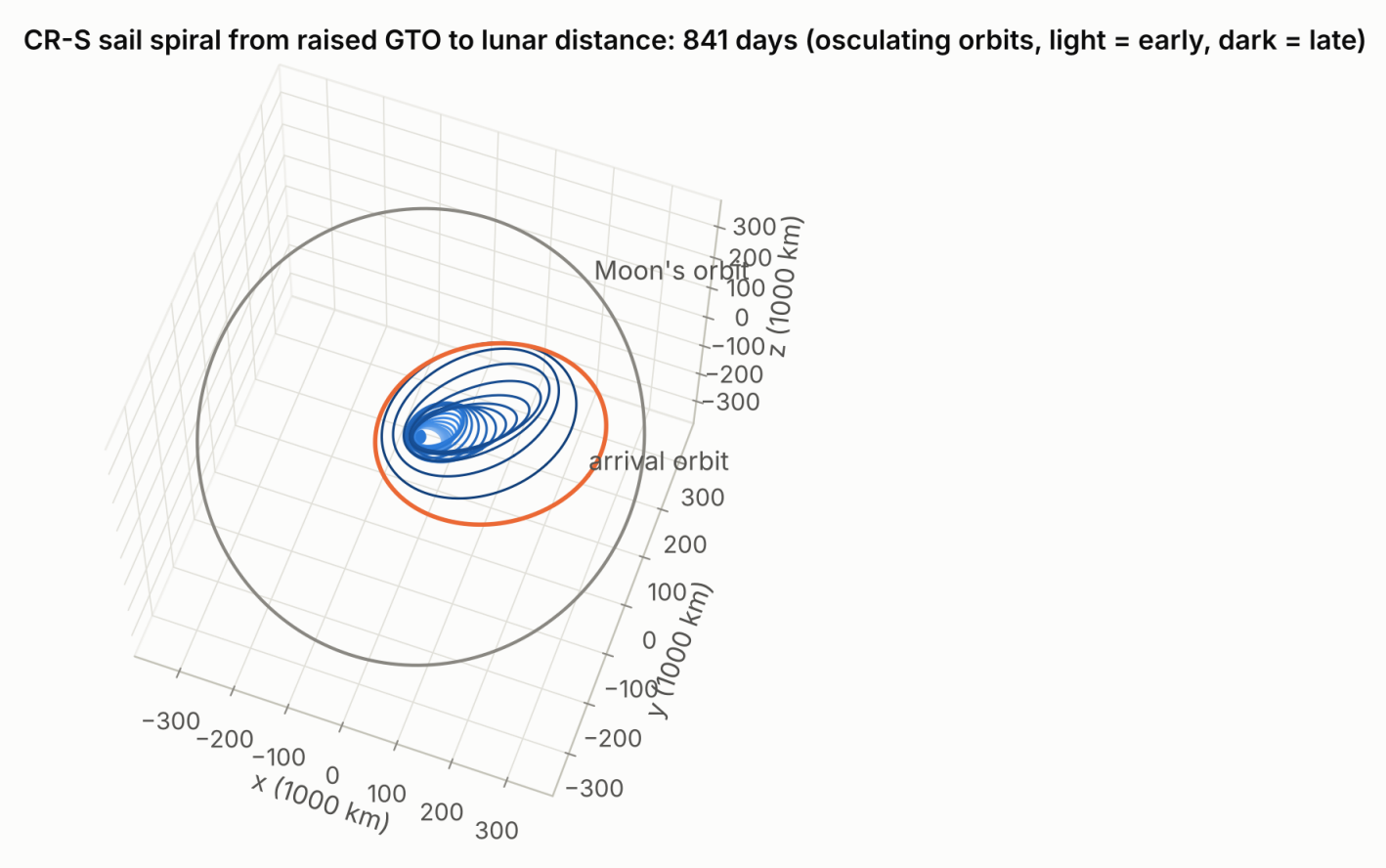
1,991 d

1,000 km LEO, 28.5 deg

TLI: lunar distance in 2.6 d, then ~89 d sail capture (~27 d hybrid).

Q1 Spiralling out from GTO

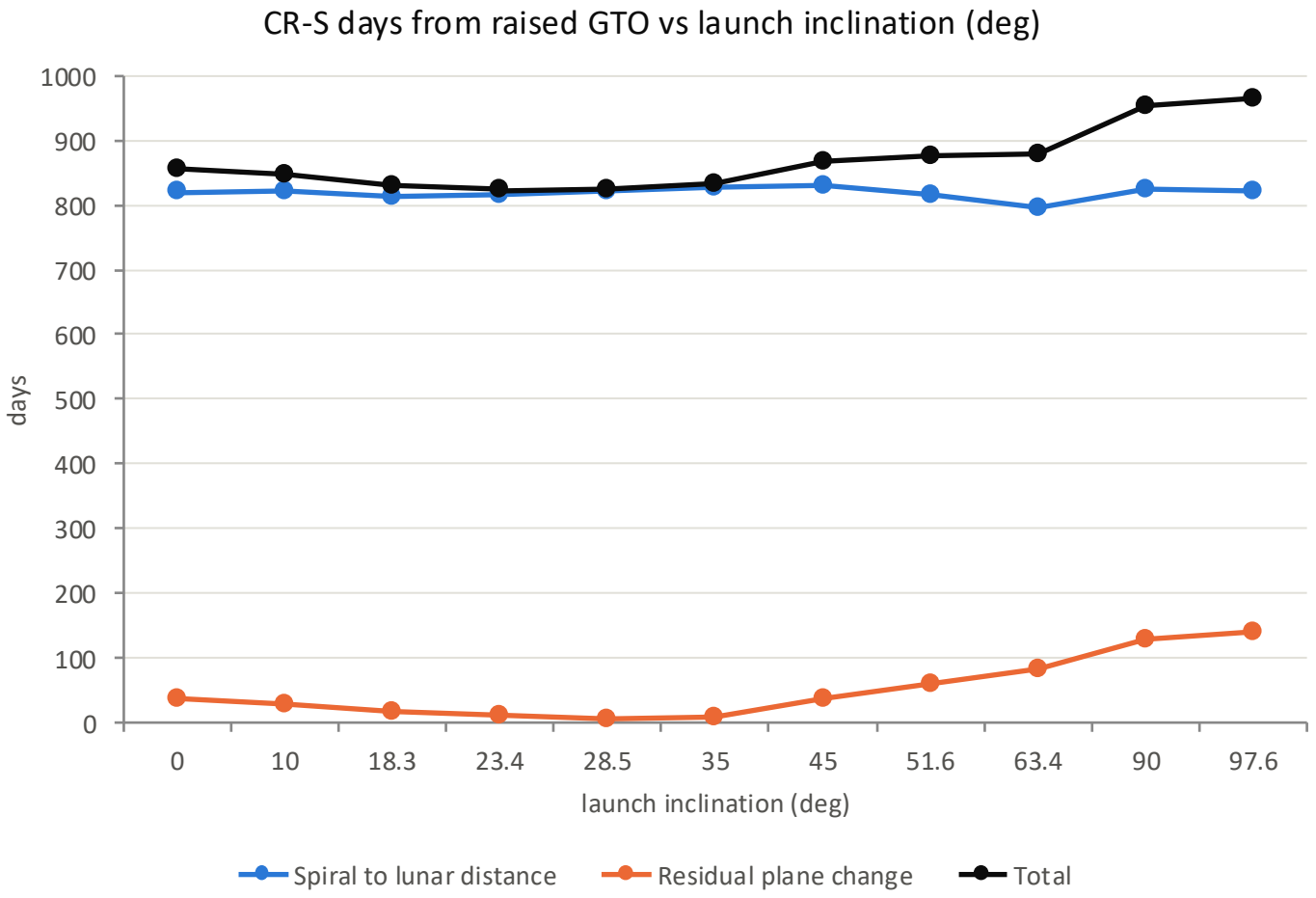
Apogee grows ~1 % per day; the arrival orbit reaches the Moon's distance after ~2.3 years

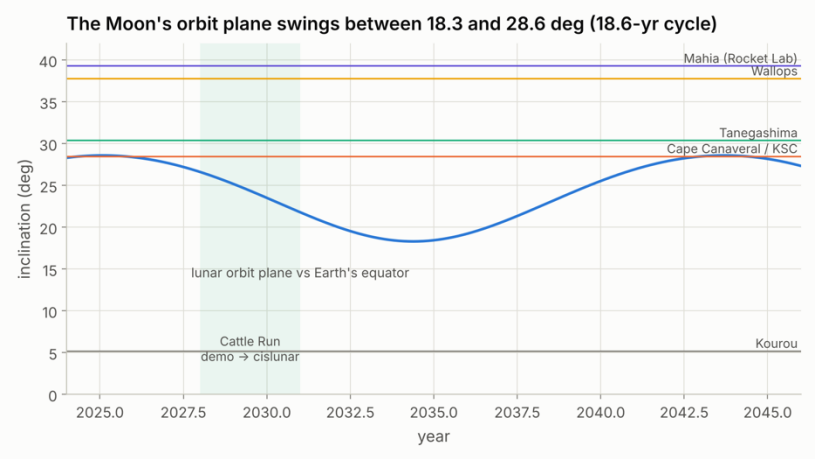


- Sail deployment in a standard 250 km-perigee GTO is survivable only if the sail stays edge-on through every 10 km/s perigee pass; then sunlight lifts perigee to ~3,450 km in 120 days.
- Robust option: raise perigee to 1,000 km first (75 m/s from the upper stage, a kick motor or SEP).
- Hybrid SEP is least valuable here: with unlimited propellant CR-H cuts GTO transit to 207 d but burns 23 kg.
- Model check: averaged vs full Cowell with Sun and Moon gravity agree within 0.1 % in apogee after 60 days.

Q2 Launch toward the lunar plane: ~28.5 deg

Spiral time is insensitive to inclination (+/-3 %); the cost is the plane mismatch at arrival

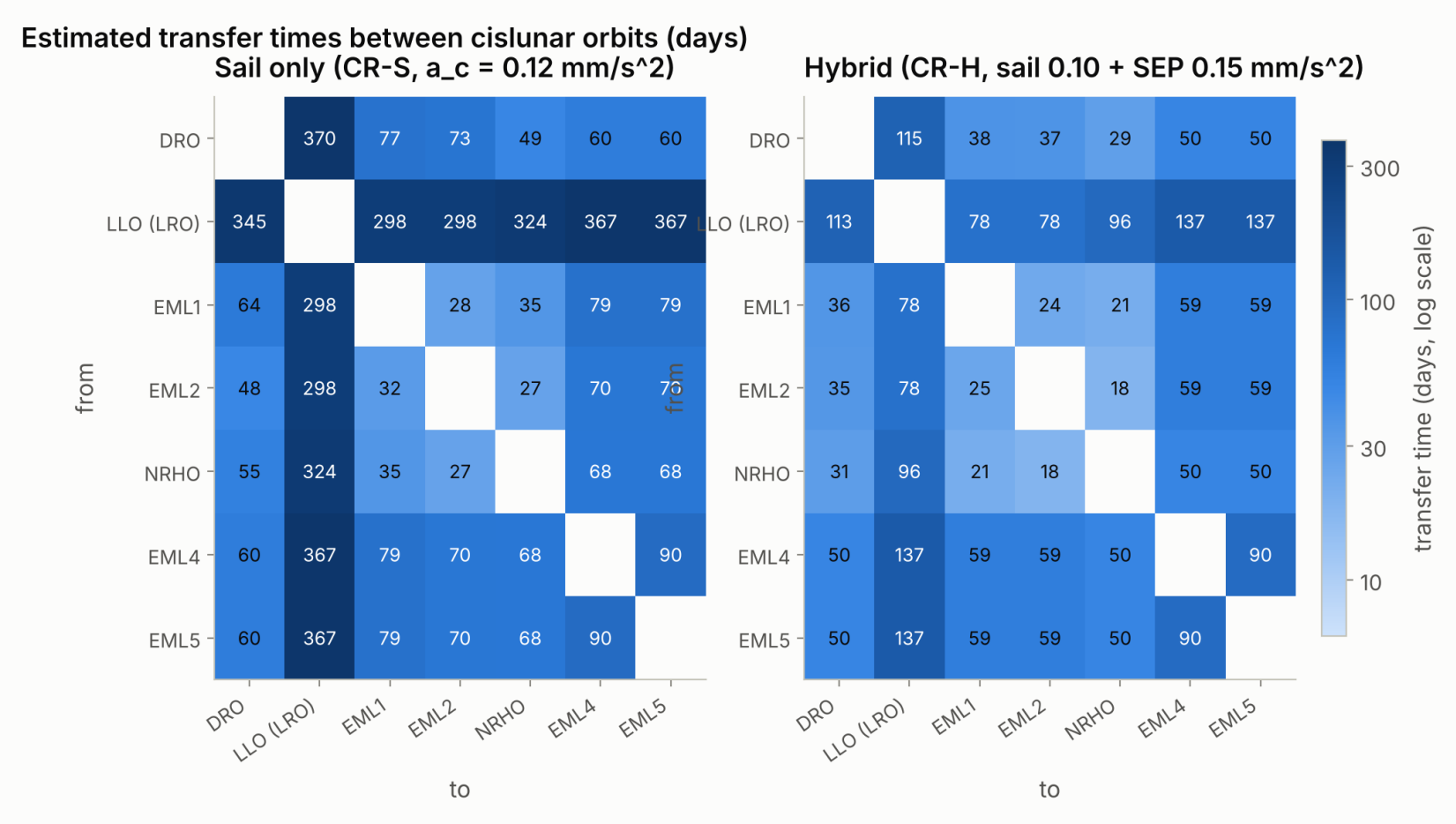




- Lunar plane: 26.6 deg (2028) -> 23.5 deg (2030) -> 18.3 deg (2034).
- Optimum 23.4-28.5 deg; SSO launches add ~139 d.
- The sail itself drifts the plane toward the ecliptic: a 28.5 deg launch arrives near 21 deg.
- LEO demo: dawn-dusk SSO is ~20 % better than noon-midnight (no eclipses).

Q3 Cislunar slots are weeks apart; LLO is a year away

Energy-change ensembles (12 start x 8 Sun phases) + natural coasts, CR3BP with rotating Sun line



27-77 d

halo / NRHO / DRO legs (sail)
hybrid 18-38 d

60-79 d

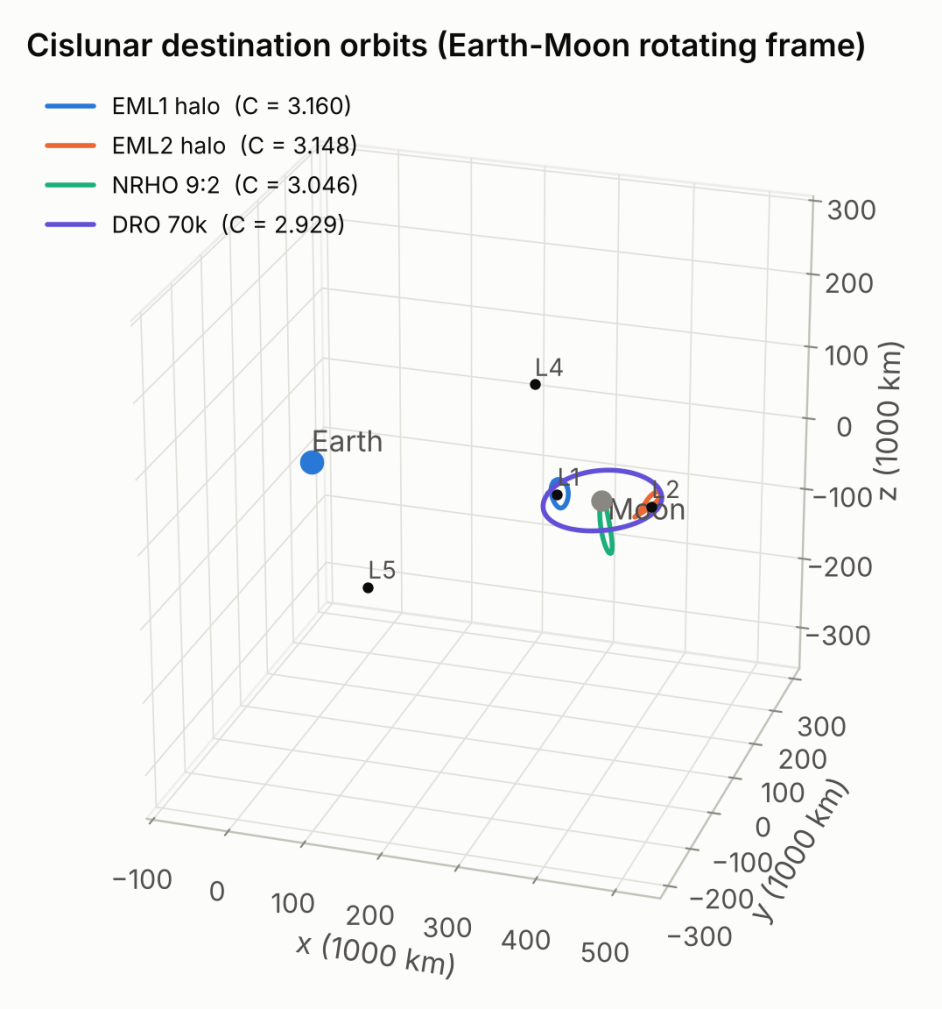
to L4 / L5 sentinel slots
(drift-dominated)

298-367 d

any leg touching 100 km LLO
hybrid 78-137 d

Q3 Destination orbits for a cislunar constellation

Computed by differential correction and continuation; the 9:2 NRHO matches Gateway's reference orbit



Orbit	Jacobi C	Period (d)	Stability
EML1 halo	3.1596	12.0	911.9
EML2 halo	3.1477	14.8	561.5
NRHO 9:2	3.0465	6.6	1.3
DRO 70k	2.9285	13.9	1.0
EML4	2.9880	-	1.0
LLO 100 km polar	5.4935	0.1	1.0

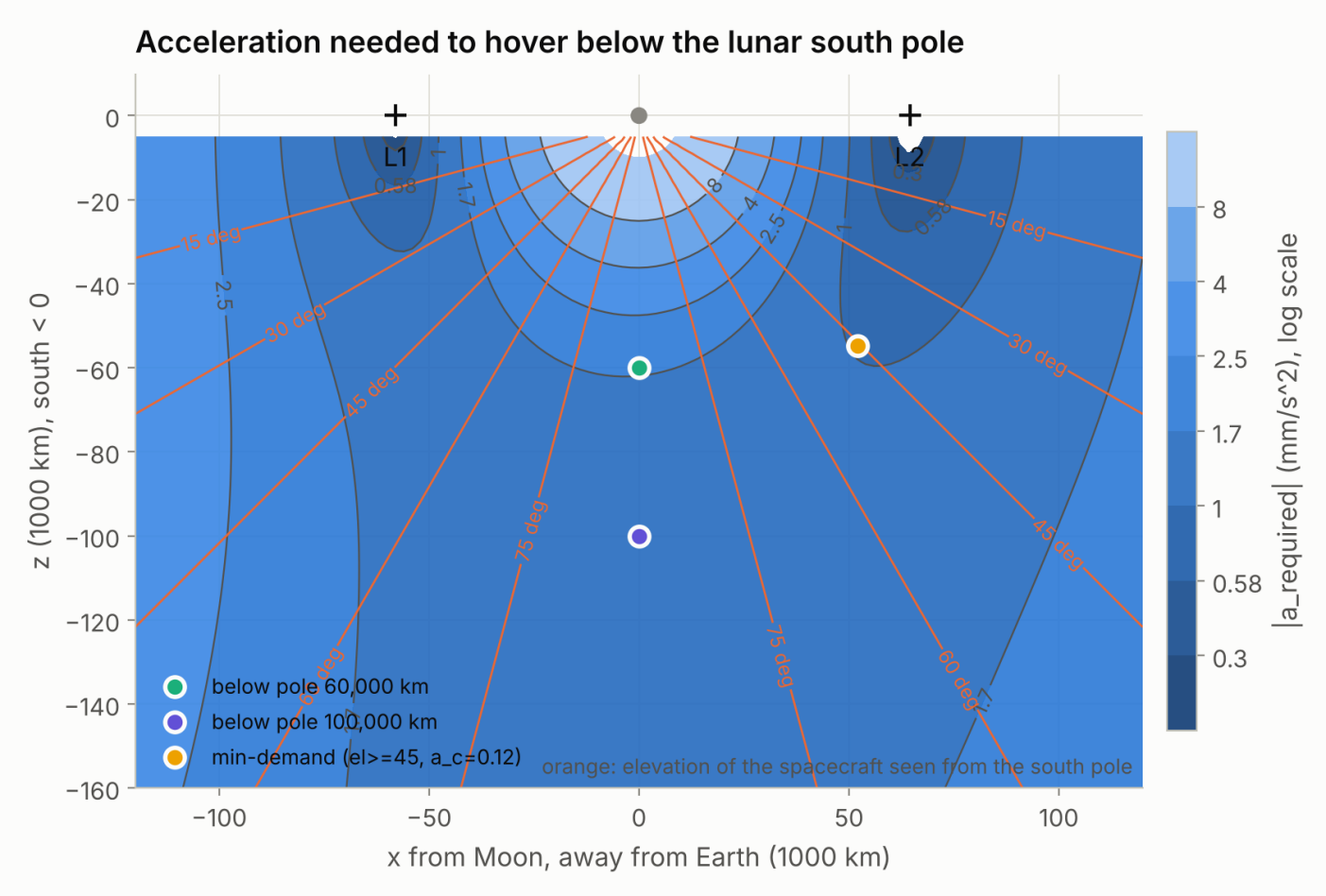
Lower C = higher energy. The DRO is neutrally stable (index 1.0) - an ideal parking slot needing almost no station keeping.

Operating there

Q4 lunar pole-sitter | Q5 in-irons limits | Q6 field of view | Q7 hybrid manoeuvrability

Q4 A lunar pole-sitter is an SEP mission

Hovering below the south pole needs 1.2-1.8 mm/s^2 continuously; the sail pushes the wrong way



1.15-1.77

mm/s^2 required, 100,000-60,000 km below the pole (10-14x CR-S a_c)

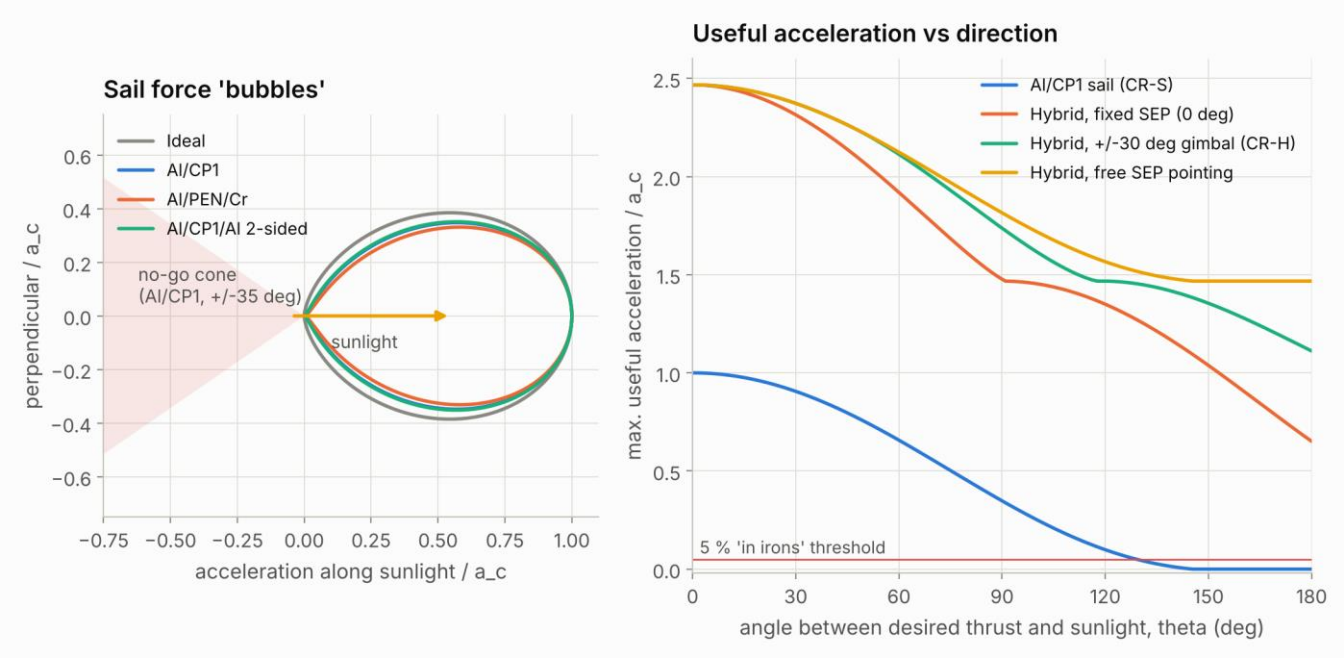
36-56 km/s

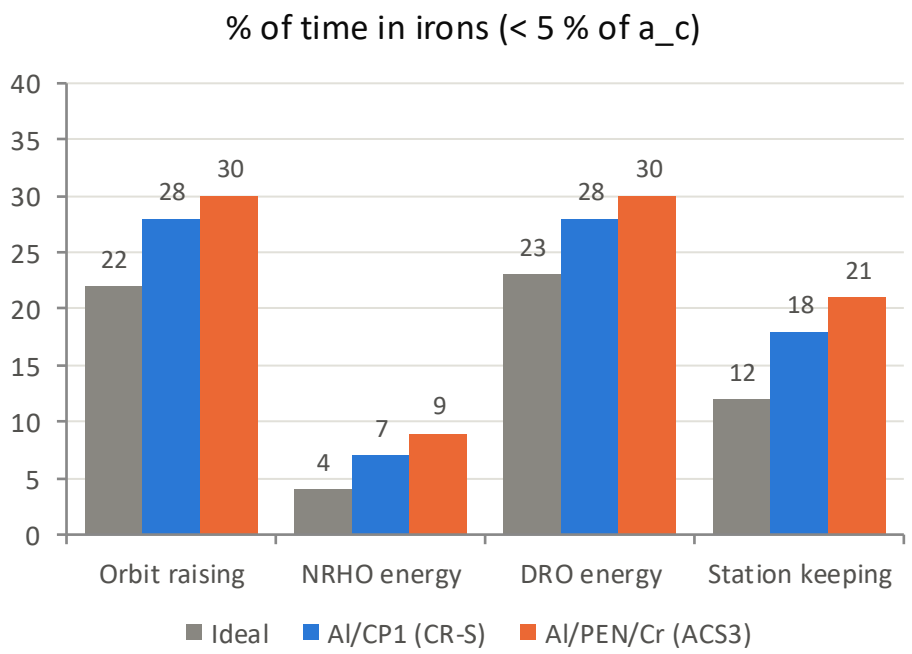
SEP delta-v per year (71 %-85 % of mass per year at Isp 3,000 s)

- A sail alone never holds a static hover (0 % of the month); even a 1.7 mm/s^2 sail removes only 10-40 % of the SEP demand.
- Sail-only coverage orbits need 0.58-1.7 mm/s^2 (Ozimek, Wawrzyniak): 5-14x CR-S.
- Recommendation: polar coverage from two NRHO / L2-halo spacecraft.

Q5 Sails are 'in irons' within ~35 deg of the Sun

A real film can tilt its force at most 50-56 deg from sunlight; any hybrid removes the dead zone





34.5 deg

no-thrust cone (Al/CP1);
50.8 deg below 5 % of a_c

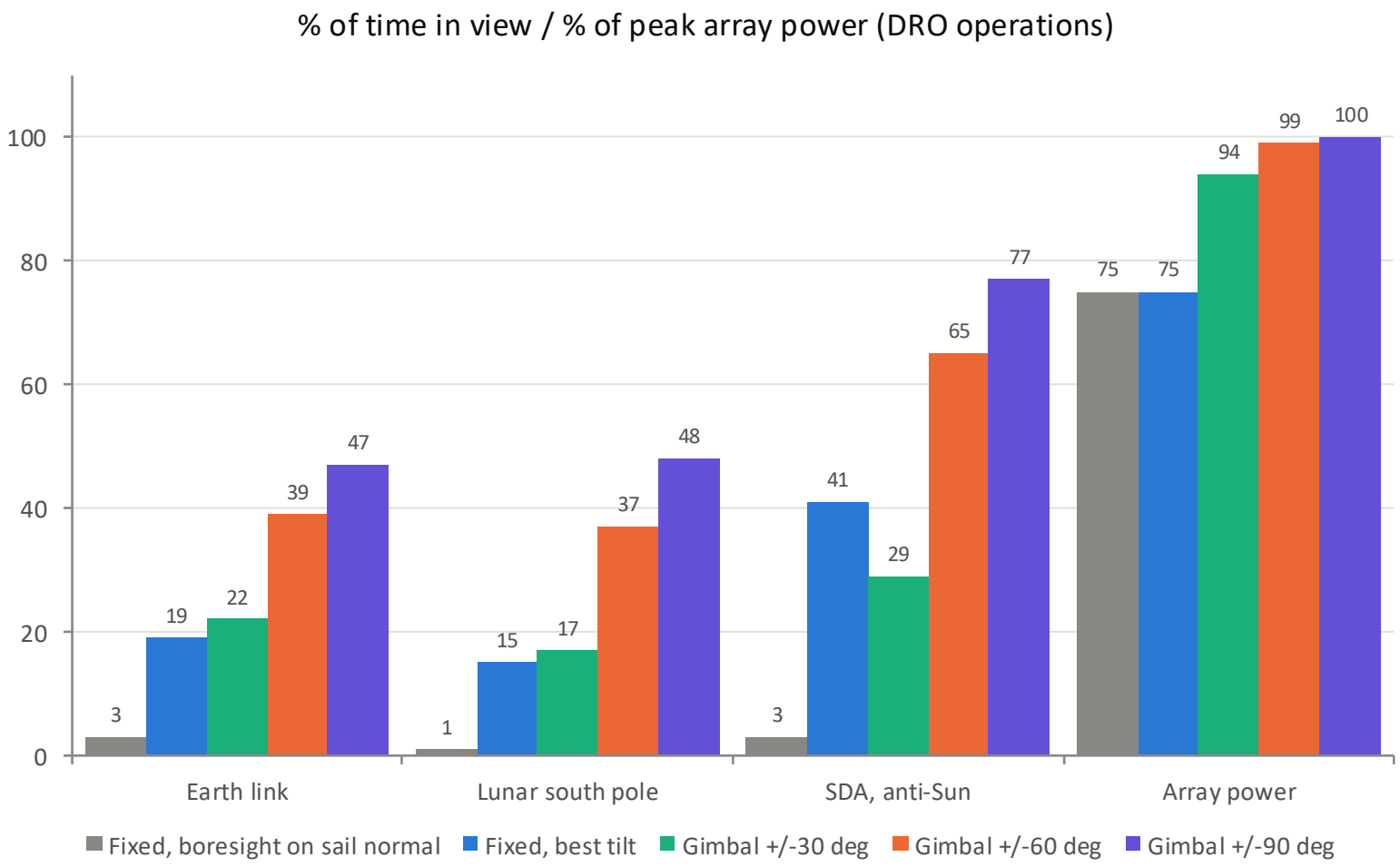
1.11 a_c

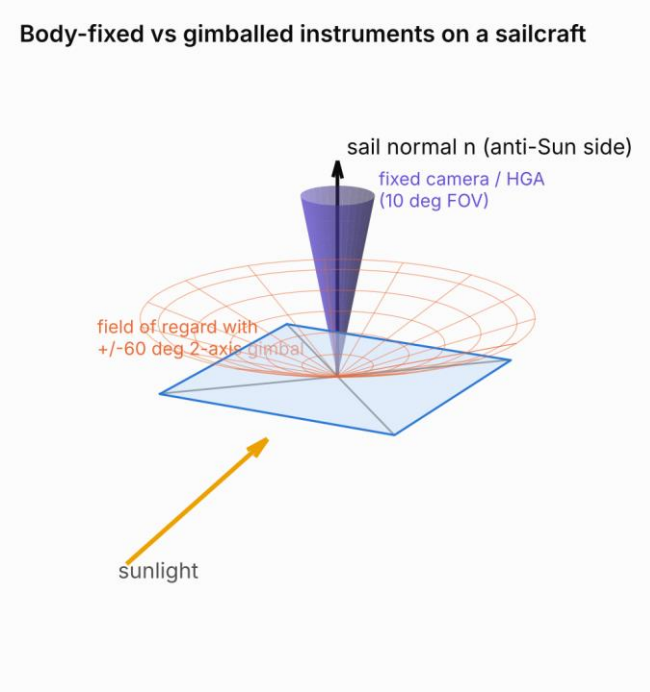
sunward thrust of CR-H
(+/-30 deg gimbal)

All hybrid variants: 0 % in irons. Even a fixed SEP thruster gives 0.65 a_c toward the Sun.

Q6 Gimbal the bus: Earth access 3 % -> 39 %

Propulsion owns the sail attitude, so body-fixed instruments point wherever thrust demands

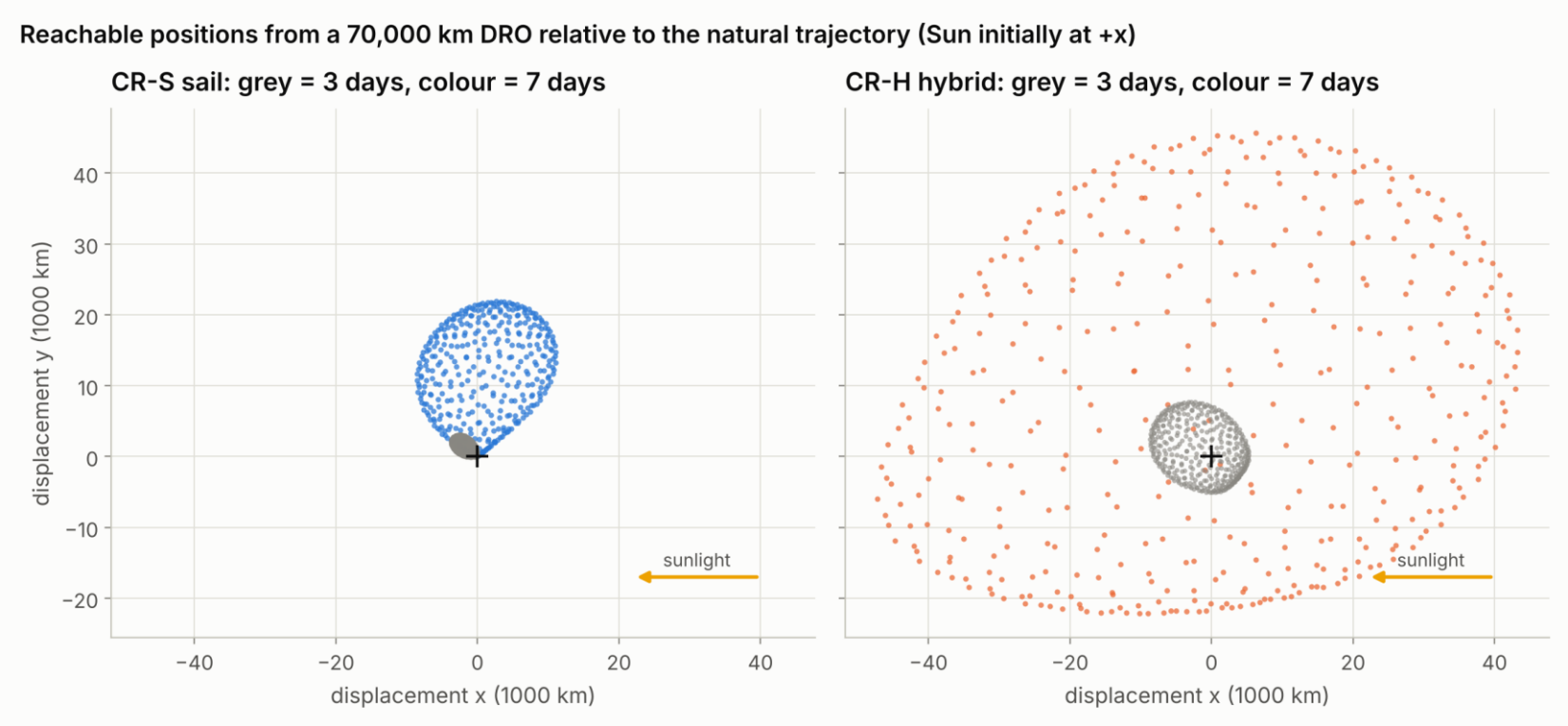




- Fixed arrays average 75 % of peak power and fall to zero when the sail feathers; +/-30 deg recovers 94 %.
- The sail blocks half the sky: carry antennas on both faces.

Q7 The hybrid reaches every direction

Constant-direction steering ensembles from a 70,000 km DRO (Sun initially at +x)



49k km

7-day reach, hybrid
(sail 22k km, lopsided)

0.45-1.00 d

10 m/s in any direction (hybrid)
sail 0.94-2.71 d, never sunward

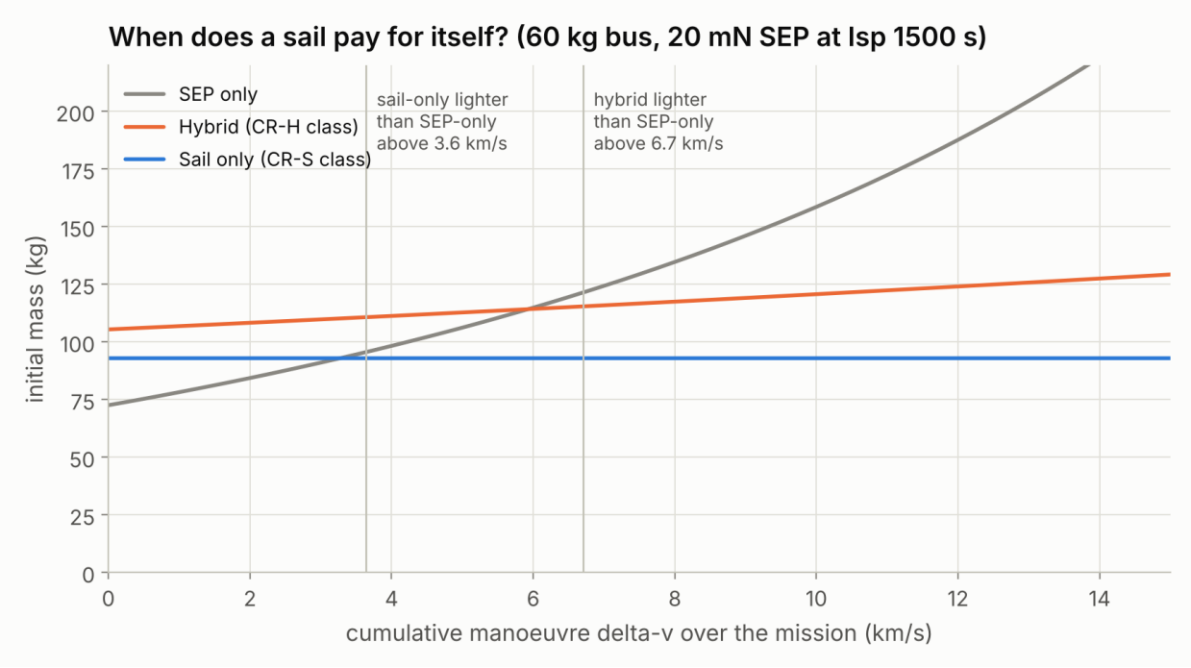
0.187

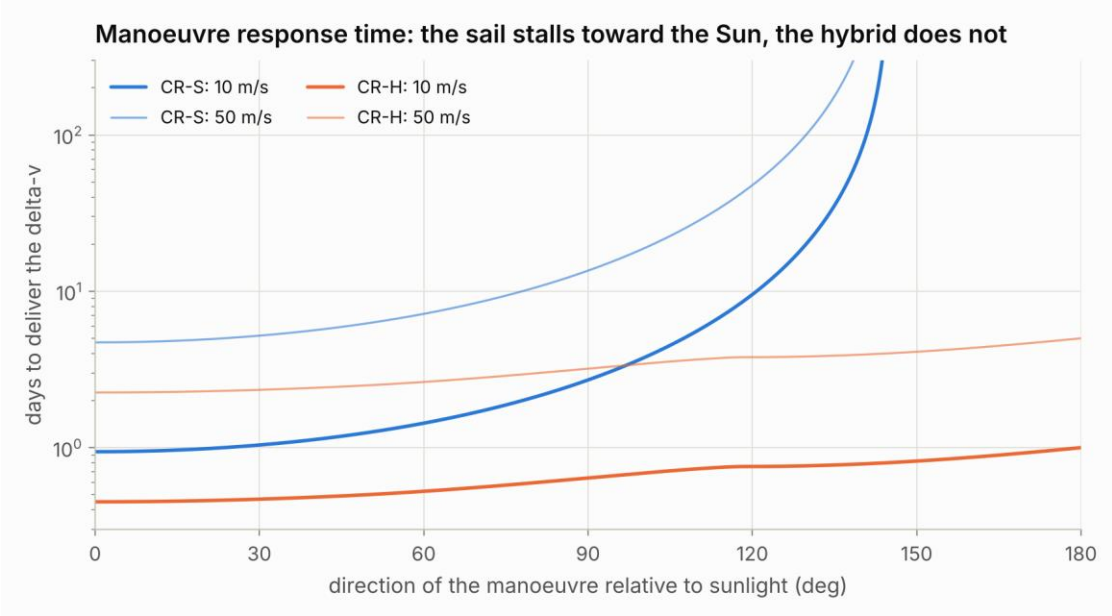
mm/s^2 mean in all directions
(sail 0.048)

Hybrid acceleration: 0.257 away from the Sun, 0.181 perpendicular, 0.116 toward it (mm/s^2).

Q7 When does a sail pay for itself?

Sails win for long, manoeuvre-intensive missions - the Cattle Run constellation use case





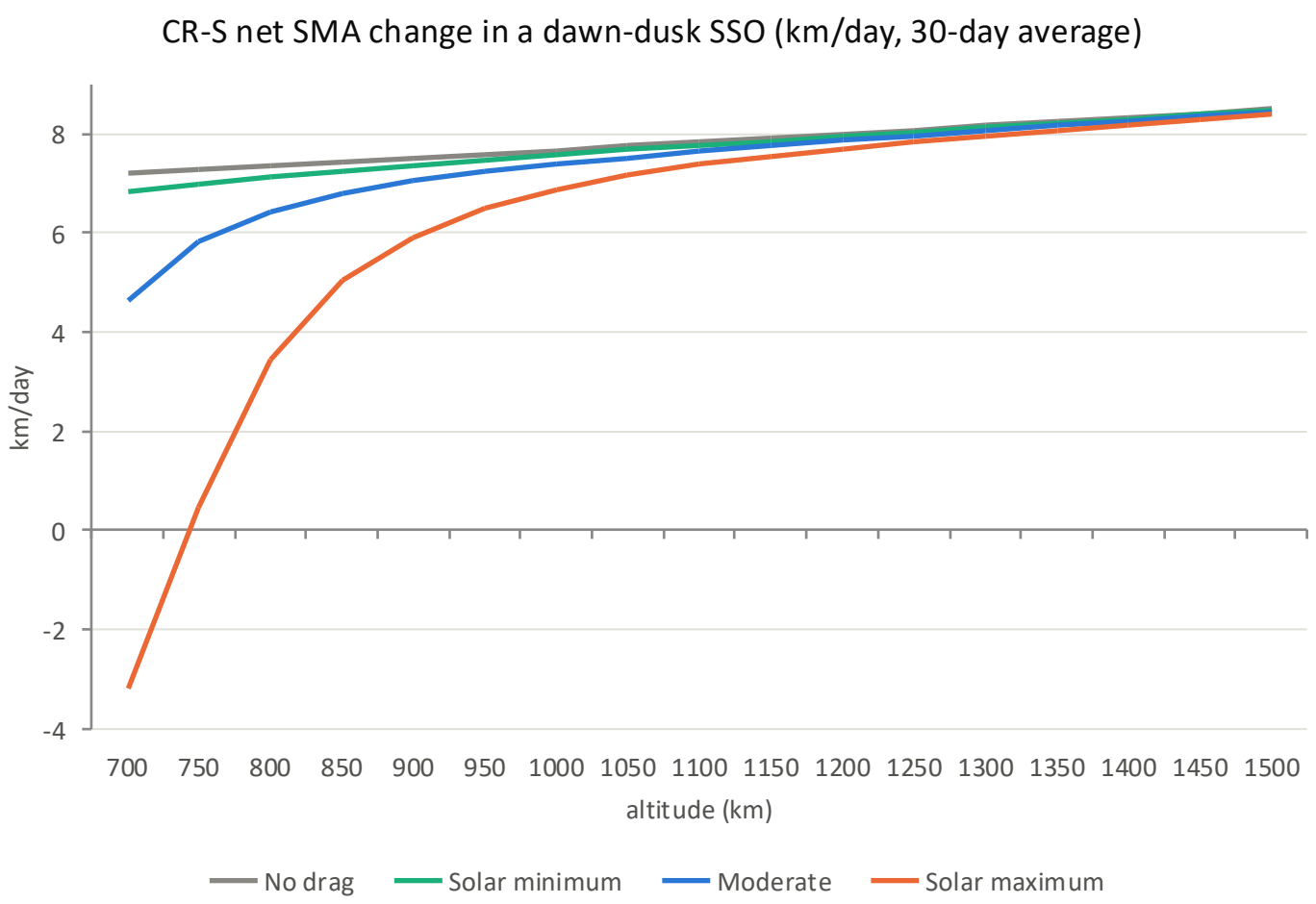
- Sail system ~33 kg: sail-only beats SEP-only after ~3.6 km/s of cumulative manoeuvring, the hybrid after ~6.7 km/s.
- Use SEP tactically (in-irons gaps, fast response, fine station keeping) and let the sail supply the bulk delta-v: CR-H's 6 kg then lasts for years.

Demonstrating it

Q8 lowest acceptable orbit for the below-GEO demonstrator

Q8 Drag sets the floor at 650-800 km

Drag and sail thrust both scale with area/mass: no sail is big enough to fly below the break-even altitude

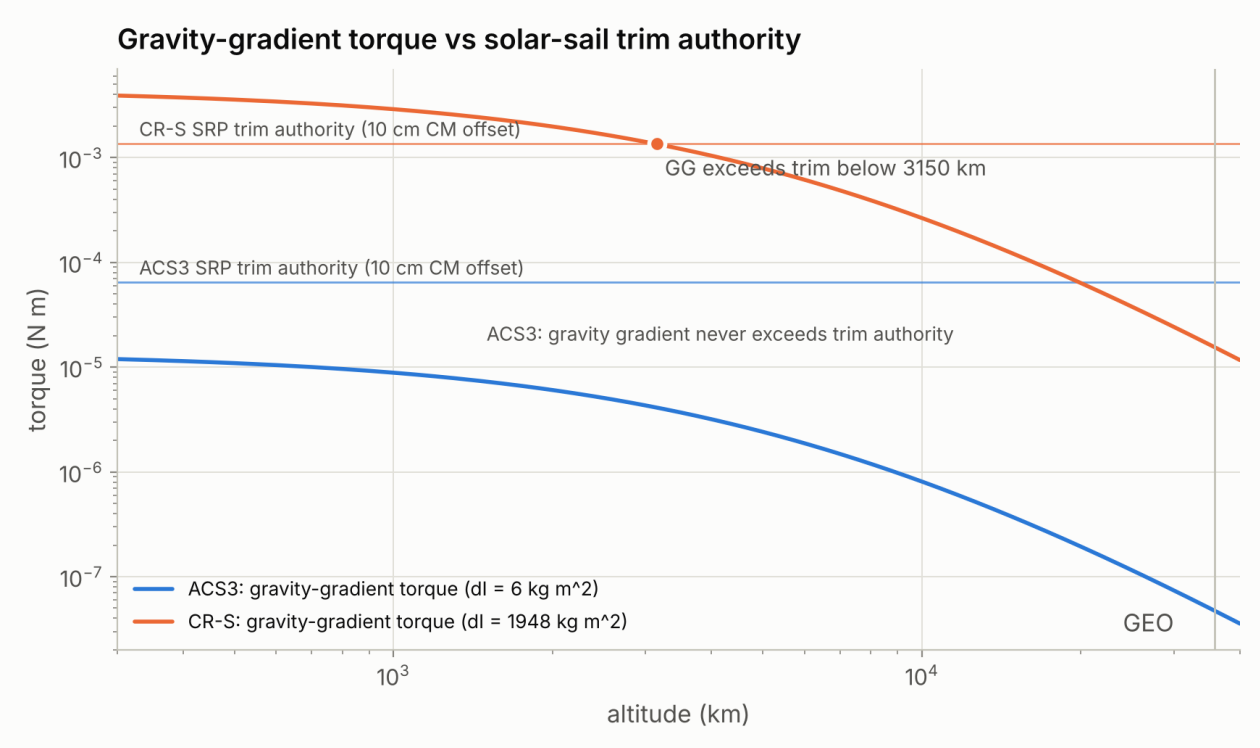


Criterion (km)	Min	Mod.	Max
Drag = thrust (orbit mean)	505	655	785
Break-even, dawn-dusk	500	632	744
Break-even, noon-midnight	506	646	754
Drag <= 25 %, dawn-dusk	550	750	900
Drag <= 25 %, noon-midnight	650	800	1,000

Cycle 25 is declining in 2028: plan for F10.7 ~ 150 with excursions above 200.

Q8 Lowest acceptable orbit: ~1,000 km dawn-dusk SSO

Robust at solar maximum, 400x less atomic oxygen than 550 km, ~220 km of raising per month for CR-S



Altitude (km)	AO erosion, ram face (um/yr)	Passive decay, CR-S
550	12.2	10 d
700	1.5	25 d
800	0.4	65 d
900	0.11	165 d
1000	0.03	365 d

- Full-size 1,653 m^2 sail: gravity gradient exceeds 10 cm CM-trim authority below ~3,150 km - size wheels/magnetorquers or demo a 100-500 m^2 sail.
- 550 km rideshare: CR-H climbs to 1,000 km with the sail stowed in 18 days using 2.1 kg.

Left: gravity-gradient torque (worst-case attitude) vs sail trim authority.

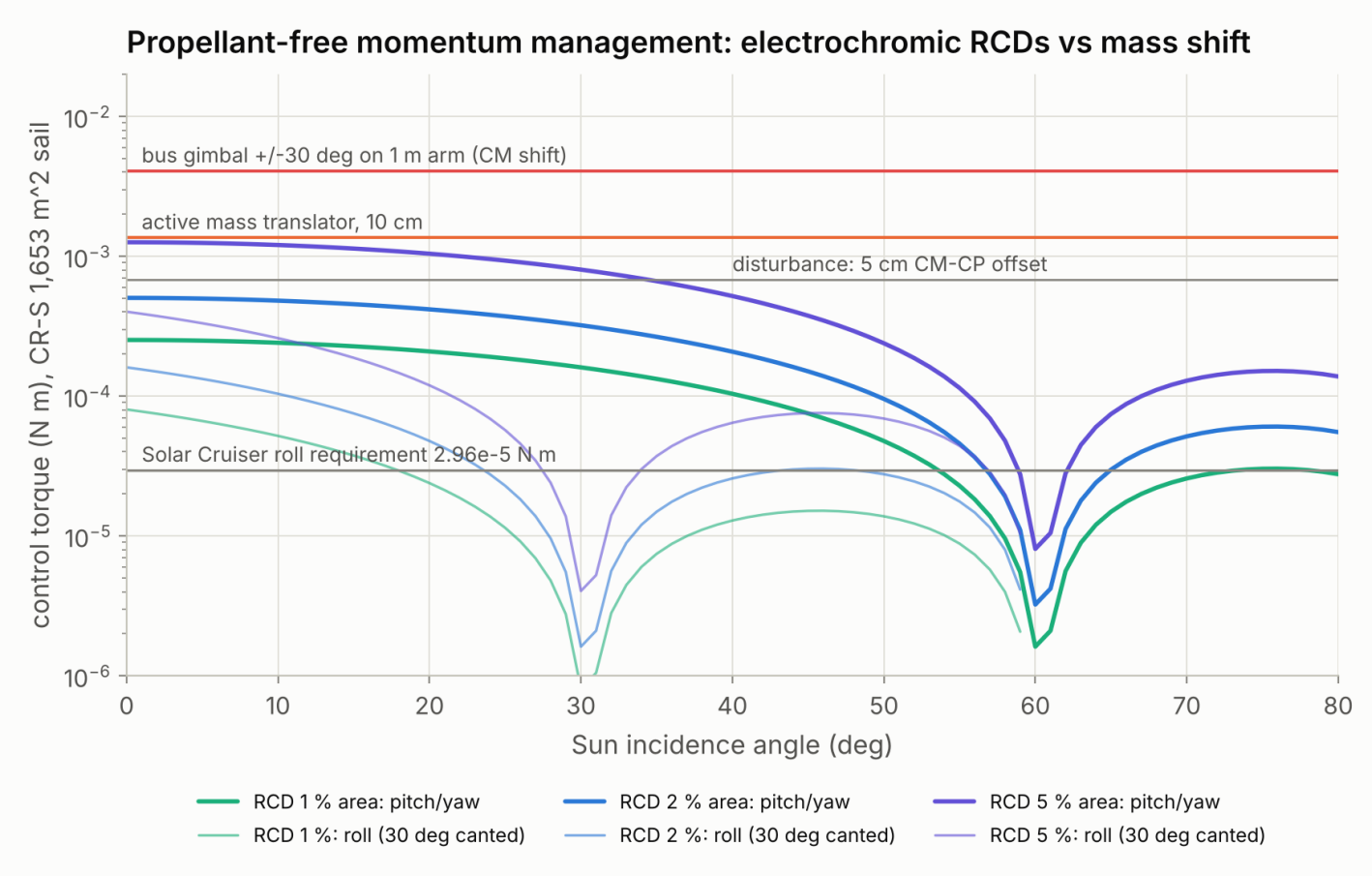
Design options scorecard

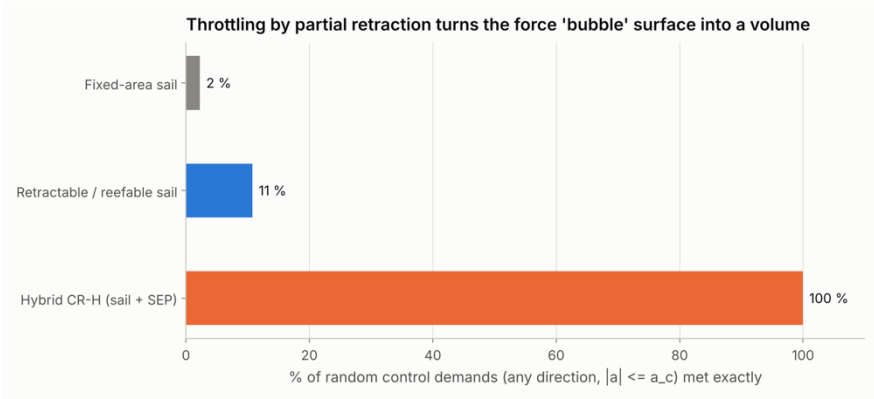
Every option in the outline was simulated; recommendations in the right-hand column

Option	Quantified benefit	Cost / risk	Call
Hybrid sail + SEP	No in-irons; 2-4x agility; SEP climb from low rideshare	+20 kg (18 %), propellant	**Adopt**
Gimballed SEP (+/-30 deg)	Sunward 1.11 vs 0.65 a_c; CM trim 4e-3 N m	Gimbal mass, plume limits	**Adopt**
Retractable sail	Throttle: 11 % vs 2 % of demands met exactly; drag/debris avoidance	+30 % deployer mass, wear	**Ground-mature, fly on demo**
Electrochromic RCD (2 % area)	5.1e-4 N m pitch/yaw, wheel-free roll	Authority nulls near 60 deg incidence	**Adopt for roll**
Double-sided film	Efficiency 0.934 vs 0.908	Runs 139 C at normal incidence	**Keep single-sided**
Spin / yo-yo deployment	~11 kg lighter sail system	Too slow for LEO steering (needs ~0.06 deg/s)	**Rigid booms**

Propellant-free attitude and momentum control

A 5 cm CM-CP offset builds 59 N m s of momentum per day on the full-size sail - trim continuously





- Bus gimbal (+/-30 deg, 1 m arm) gives ~3x the authority of a 10 cm mass translator.
- RCD panels at 2-5 % of area match the disturbance at low incidence and meet Solar Cruiser's roll requirement.
- Partial retraction throttles the sail: 5x more control demands met exactly.

Questions the proposal should also answer

Surfaced by the analyses; ordered by priority for the Cattle Run proposal

A. Film and coating life at the demo altitude and through the radiation belts

B. Collision and conjunction risk of a 1,653 m^2 sail at ~1,000 km; screening with continuous thrust

C. Orbit determination with attitude-dependent SRP; force-model calibration in flight

D. Autonomy architecture: safe feathering, momentum management, fault recovery

E. LEO gravity-gradient and aerodynamic torques for a large sail

F. Sail billow, wrinkles and boom thermal deflection effects on force and torque

G. Survival of lunar and Earth eclipses (thermal snap, boom contraction loads)

H. Constellation deployment and phasing from a single launch

I. Antenna placement when the sail blocks half the sky

J. SEP plume impingement on the film and resulting gimbal limits

K. Station-keeping cost in the full ephemeris model with solar gravity

L. End of life: drag-sail disposal in LEO, heliocentric escape from cislunar space

Recommended concept and next steps

A hybrid sailcraft, demonstrated at ~1,000 km and deployed to DRO, NRHO, L1/L2 and L4/L5 slots

Architecture (CR-H class)

Solar Cruiser-heritage 1,653 m^2 single-sided CP1 sail on rigid booms; RCD roll control; two-axis gimballed bus (+/-30-60 deg) with a 15-25 mN Hall thruster, arrays and instruments; antennas on both faces.

Phase 1 demonstrator

~1,000 km dawn-dusk SSO (or 550 km rideshare + 18-day SEP climb). Success: measured a_c within 10 %, ~5-7 km/day raising, autonomous steering and momentum management, safe feathering.

Phase 2 cislunar operations

Launch toward the lunar plane (28.5 deg) to GTO/GEO or TLI; slots in DRO, NRHO and L1/L2 halos with L4/L5 sentinels; polar coverage by paired NRHO spacecraft.

Simulator next steps

Trajectory optimisation for transfers, full-ephemeris / BCR4BP station keeping with a sail-aware controller, slew-limited attitude dynamics, sail-shape uncertainty, GPU Monte Carlo of autonomy faults.

Backup: assumptions and reproduction

All results are reproducible from the repository

- Flat sail, nominal optics, no degradation (real performance typically 5-15 % lower)
- Instantaneous attitude changes in trajectory runs; agility treated separately
- Averaged Earth-orbit dynamics without lunisolar perturbations (0.1 % after 60 days)
- CR3BP with rotating Sun line; transfer times = energy change + natural coast (+/-30 %)
- NRLMSIS 2.x equatorial densities, C_D 2.2, 1 % edge-on residual area
- Mass model calibrated on ACS3 and Solar Cruiser; SEP sizing after Ceriotti & McInnes

Command	Effect
`python simulator/run_all.py`	All analyses -> results/*.json, figures/*.png
`CISLUNAR_BACKEND=torch python simulator/run_all.py`	Same on PyTorch (CUDA GPU when present)
`python reports/build_report.py`	Rebuild the PDF report
`node reports/build_deck.js`	Rebuild this deck